



May 2025

ME1003-2 – ELEMENTS OF MECHANICAL ENGINEERING
(For CV)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- The heating of dry and saturated steam above saturation temperature is known as**
A) Enthalpy
B) Superheating
C) Super saturation
D) Latent heat
- The property which permits a material to be drawn out longitudinally to a reduced section under the action of tensile force**
A) Ductility
B) Elasticity
C) Malleability
D) Plasticity
- The slope of the stress-strain curve in the elastic deformation region is**
A) Plastic modulus
B) Elastic modulus
C) Poisson's ratio
D) Rigidity Modulus
- The energy stored in the earth's crust is called**
A) Oceanic thermal energy
B) Biomass energy
C) Geo-thermal energy
D) Potential energy
- Function of an air compressor is to,**
A) Discharge high pressure water
B) Suction of fuel in engines
C) Pressurize air and deliver
D) Compress the oil
- The output of the nozzle is**
A) Low pressure and high velocity steam
B) Low pressure and low velocity steam
C) High pressure and high velocity steam
D) Very high pressure and high velocity steam
- Which type of water turbine operates with a high head of water and directs the water onto buckets to generate power?**
A) Pelton wheel
B) Kaplan turbine
C) Francis turbine
D) Parsons turbine
- In a gas turbine, which type of cycle is used where air is compressed and mixed with fuel before ignition?**
A) Open cycle
B) Closed cycle
C) Rankine cycle
D) Otto cycle
- The function of flywheel is to,**
A) Reduce efficiency of engine
B) Store and release energy during work cycle
C) Regulate the speed
D) Increase weight of the engine
- The expansion device in any refrigeration system is connected between**
A) Evaporator and Condensor
B) Evaporator and Compressor
C) Compressor and Condensor
D) Evaporator and Absorber
- What is the main differentiator between a hybrid vehicle and a fully electric vehicle?**
A) Battery capacity
B) Engine power
C) Fuel type
D) The presence of an internal combustion engine

12. In a 4-stroke diesel engine, how is ignition initiated to start the combustion process?
 - A) Spark ignition
 - B) Compression ignition
 - C) Fuel injection
 - D) External ignition source
13. Electronic components are joined by which of the following methods?
 - A) Spot Welding
 - B) Seam Welding
 - C) Soldering
 - D) Brazing
14. Type of oxy-acetylene flame which has highest temperature,
 - A) Carburizing
 - B) Oxidizing
 - C) Neutral
 - D) Lower-neutral
15. The teeth on the spur gears with respect to axes of gears are
 - A) Parallel
 - B) Perpendicular
 - C) Inclined
 - D) Intersecting
16. In cross belt drive the driver and driven pulley always rotates in
 - A) Same direction
 - B) Opposite direction
 - C) Any direction
 - D) None of above
17. Which one of the following engineering deals with controlling and intelligence of robots?
 - A) Electrical
 - B) Mechanical
 - C) Computer
 - D) Architecture
18. The robot designed with polar configuration has
 - A) Three linear movements
 - B) Three rotational movements
 - C) Two rotational and one linear movement
 - D) Two linear and one rotational movement
19. Boring is an operation in which the drilled hole diameter is,
 - A) Enlarged
 - B) Reduced
 - C) Finished
 - D) Texturized
20. Which of the following automation has high production rate.
 - A) Flexible automation
 - B) Programmable automation
 - C) Fixed automaton
 - D) Linear Automation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List the mechanical properties of materials. Explain any three of them in detail.	5	L*1	1	1
	b) Define the following. (i) Enthalpy (ii) Radiation (iii) Poissons Ratio	3	L1	1	1
	c) Define the meaning of prime mover. Explain the working of Francis reaction water turbine with a suitable figure.	8	L2	2	1
2.	a) Explain the process formation of steam with suitable sketches and highlighting various terminologies involved.	8	L1	1	1
	b) What is the meaning of an air compressor. Explain the working of a single stage air compressor with help of a suitable figure.	8	L2	2	1
3.	a) What is the meaning power plant. Explain the working of a Solar power plant with the help of a sketch.	8	L2	1	1
	b) Describe the working of Reaction Steam turbine (Parsons turbine) with the help of suitable diagrams.	8	L2	2	1
Unit – II					
4.	a) The following observations were made during a trial on a 4-stroke diesel engine. Cylinder diameter = 25 cm; Stroke of the piston = 40 cm; Crankshaft speed = 250 rpm; Brake load = 70 kg; Brake drum diameter = 2 m; Mean effective pressure = 6 bar; Diesel oil consumption = 0.1 m ³ /min; Specific gravity of diesel = 0.78; Calorific value of petrol = 43,900 kJ/kg. Find: (i) Indicated power (ii) Brake power (iii) Mechanical Efficiency (iv) Indicated Thermal efficiency.	8	L2	3	1
	b) Explain the principle and process of Gas welding with suitable diagrams.	8	L1	4	1

5.	a)	Explain the working of a 4-stroke diesel engine with necessary figures indicating all strokes and P-V diagram.	8	L3	3	1
	b)	Define Gear. List different types of gears and Explain any Three with suitable sketches.	8	L1	4	1
6.	a)	Explain the working of electric and hybrid vehicles with the help of block diagrams.	8	L2	3	1
	b)	Define welding. Explain the process of MIG welding with a suitable sketch.	8	L2	4	1
Unit – III						
7.	a)	Explain different Types of Automation with suitable examples.	8	L1	5	1
	b)	Explain with figures the following Milling operations: Plane Milling, End Milling and Slot Milling.	6	L1	5	1
	c)	Define of Robot and Robotics.	2	L1	5	1
8.	a)	Explain with figures the following lathe machine operations: Plain Turning, Facing, Knurling, Thread Cutting.	8	L1	5	1
	b)	Explain the meaning of Automation. Explain the necessity of automation.	4	L1	5	1
	c)	List any eight Applications of Robots.	4	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
